

Digital Control System Analysis and Design

Chap. 1 Introduction

Chap. 2 Discrete Time Systems and the z-Transform

Chap. 3 Sampling and Reconstruction

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Chap. 5 Closed-Loop Systems

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Chap. 7 Stability Analysis Techniques

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4.2 The Relationship Between $E(z)$ & $E^*(s)$

Defined: $z[\{e(k)\}] = E(z) = e(0) + e(1)z^{-1} + e(2)z^{-2} + \dots$

then $E^*(s) = e(0) + e(T)\varepsilon^{-Ts} + e(2T)\varepsilon^{-2Ts} + \dots$

$$E(z) = E^*(s) \big|_{\varepsilon^{sT} = z} \quad (4-3)$$

$E(z)$ can be calculated

$$E(z) = \sum_{\substack{\text{at poles} \\ \text{of } E(\lambda)}} \left[\text{residues of } E(\lambda) \frac{1}{1 - z^{-1} \varepsilon^{T\lambda}} \right]$$

Example 4.1 vs. Example 3.3

Example 4.1

Let

$$E(s) = \frac{1}{(s + 1)(s + 2)}$$

Then, from the z-transform tables in Appendix VIII,

$$E(z) = E^*(s)|_{\epsilon^{sT} = z} = \frac{z(\epsilon^{-T} - \epsilon^{-2T})}{(z - \epsilon^{-T})(z - \epsilon^{-2T})}$$

Hence

$$E^*(s) = \frac{\epsilon^{Ts}(\epsilon^{-T} - \epsilon^{-2T})}{(\epsilon^{Ts} - \epsilon^{-T})(\epsilon^{Ts} - \epsilon^{-2T})}$$

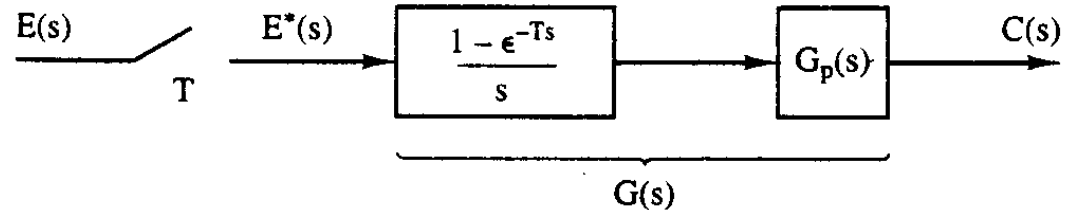
Compared with Example 3.3 via (3-10)

➡ The 4th method to evaluate $E^*(s)$.
i.e. Eq (3-7), (3-10), (3-11), and (4-3)

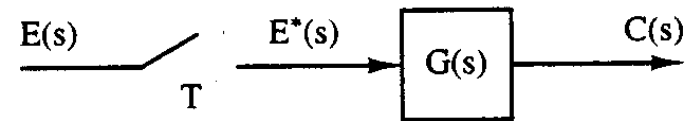
4.3 The Pulse Transfer Function

$$G(s) = \frac{1 - e^{-Ts}}{s} G_p(s)$$

$$C(s) = G(s)E^*(s)$$



(a)



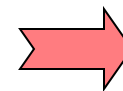
(b)

Figure 4-1 Open-loop sampled-data system.

$$C^*(s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} C(s + jn\omega_s) = [G(s)E^*(s)]^*$$

$$C^*(s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} G(s + jn\omega_s) \underline{E^*(s + jn\omega_s)}$$

$$C^*(s) = \underline{E^*(s)} \frac{1}{T} \sum_{n=-\infty}^{\infty} G(s + jn\omega_s) = E^*(s)G^*(s)$$



$$C(z) = E(z)G(z)$$

$G(z)$ is the **pulse TF**: I/O relation *at the sampling instants*. 4/34

The derivation above is completely general. Thus given any function that can be expressed as

$$\underline{A(s) = B(s)F^*(s)} \quad (4-10)$$

where $F^*(s)$ must be expressible as

$$F^*(s) = f_0 + f_1 e^{-Ts} + f_2 e^{-2Ts} + \dots$$

then, from the preceding development,

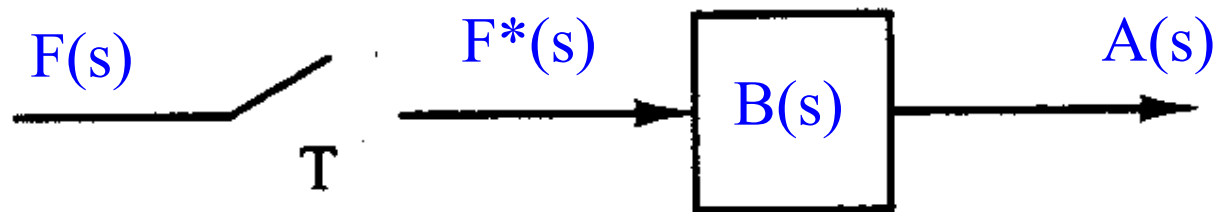
$$\underline{A^*(s) = B^*(s)F^*(s)} \quad (4-11)$$

Hence, from (4-3),

$$\underline{A(z) = B(z)F(z)} \quad (4-12)$$

where $B(s)$ is a function of s and $F^*(s)$ is a function of e^{Ts} ; that is, in $F^*(s)$, s appears only in the form e^{Ts} . Then, in (4-12),

$$\star \quad \underline{B(z) = \mathcal{Z}[B(s)], \quad F(z) = F^*(s)|_{e^{Ts} = z}} \quad (4-13)$$



Example 4.2

Suppose that we wish to find the z -transform of

$$A(s) = \frac{1 - \epsilon^{-Ts}}{s(s+1)} = \frac{1}{s(s+1)} (1 - \epsilon^{-Ts})$$

From (4-10), we consider

$$B(s) = \frac{1}{s(s+1)}$$

and

$$\underline{F^*(s) = 1 - \epsilon^{-Ts} \Rightarrow F(z) = 1 - z^{-1} = \frac{z-1}{z}}$$

Then, from the z -transform tables,

$$B(z) = \mathcal{Z}\left[\frac{1}{s(s+1)}\right] = \frac{(1 - \epsilon^{-T})z}{(z-1)(z - \epsilon^{-T})}$$

and

$$A(z) = B(z)F(z) = \frac{(1 - \epsilon^{-T})z}{(z-1)(z - \epsilon^{-T})} \left[\frac{z-1}{z} \right] = \frac{1 - \epsilon^{-T}}{z - \epsilon^{-T}}$$

Example 4.3

Given the system shown in Figure 4-2, with input $e(t)$ a unit step function, let us determine the output function $C(z)$. Now,

$$C(s) = \frac{1 - e^{-Ts}}{s(s+1)} E^*(s) = G(s)E^*(s)$$

In Example 4.2 it was shown that

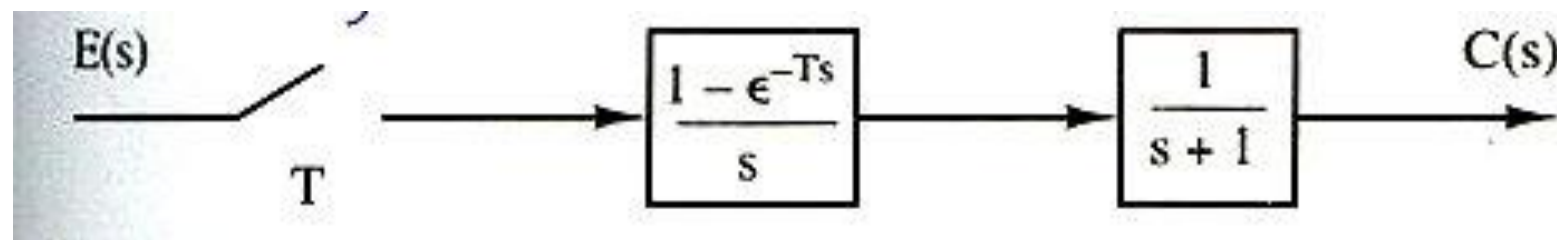
$$G(z) = \mathcal{Z}\left[\frac{1 - e^{-Ts}}{s(s+1)}\right] = \frac{1 - e^{-T}}{z - e^{-T}}$$

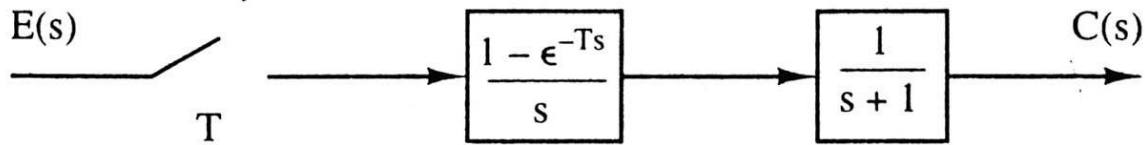
In addition, from the table in Appendix VIII,

$$E(z) = \mathcal{Z}[u(t)] = \frac{z}{z-1}$$

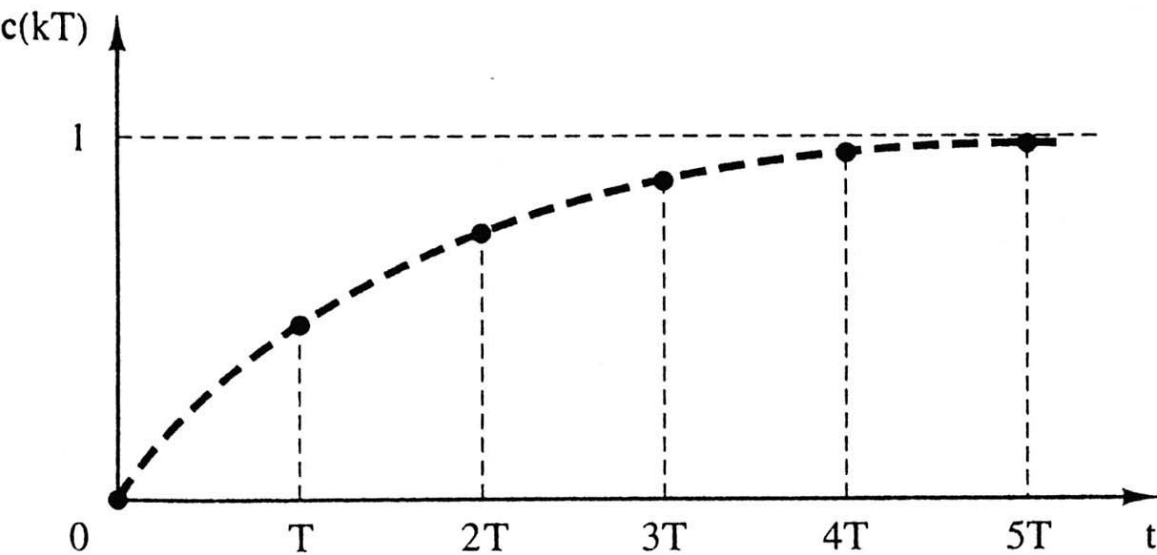
Thus

$$\underline{C(z) = G(z)E(z)} = \frac{z(1 - e^{-T})}{(z-1)(z - e^{-T})} = \frac{z}{z-1} - \frac{z}{z - e^{-T}}$$





(a)



(b)

Figure 4-2 Sampled-data system.

$c(t) \rightarrow c(nT)$: OK (sampling)

$c(nT) \rightarrow c(t)$: NG (need the data reconstruction)

dc gain

$$\begin{aligned} c_{ss}(k) &= \lim_{z \rightarrow 1} (z - 1)C(z) = \lim_{z \rightarrow 1} (z - 1)G(z)\underline{E(z)} \\ &= \lim_{z \rightarrow 1} (z - 1)G(z)\underline{\frac{z}{z - 1}} = \lim_{z \rightarrow 1} G(z) = G(1) \end{aligned}$$

A constant input of unity

assuming the $c_{ss}(k)$ exists. Since the steady-state input is unity, we see that the dc gain is given by

$$\text{dc gain} = G(z)|_{z=1} = G(1)$$

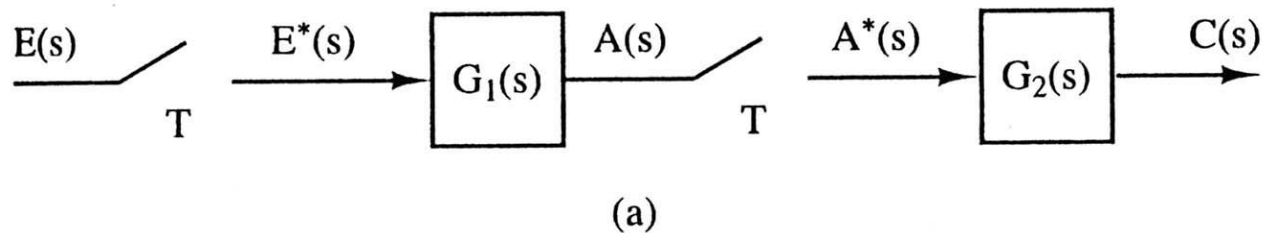
For the example above, $G(1) = 1$, which checks the time-response calculation. Since, for a constant input, the gain of the sampler/zero-order-hold is unity, the dc gain of the system of Figure 4-1 is also given by

$$\text{dc gain} = \lim_{s \rightarrow 0} G_p(s) \quad \begin{array}{l} \text{From S-domain viewpoint} \\ \text{(w/o sampler and ZOH)} \end{array}$$

Thus

$$\boxed{\text{dc gain} = \lim_{z \rightarrow 1} G(z) = \lim_{s \rightarrow 0} G_p(s)} \quad (4-14)$$

Fig. 4.3 (a)



$$C(s) = G_2(s)A^*(s)$$

and thus

$$C(z) = G_2(z)A(z) \quad (4-15)$$

Also,

$$A(s) = G_1(s)E^*(s)$$

and thus

$$A(z) = G_1(z)E(z) \quad (4-16)$$

Then, from (4-15) and (4-16),

$$\underline{C(z) = G_1(z)G_2(z)E(z)} \quad (4-17)$$

Fig. 4.3 (b)



(b)

$$C(s) = G_1(s)G_2(s)E^*(s)$$

and

$$\underline{C(z) = \overline{G_1 G_2}(z)E(z)}$$

where

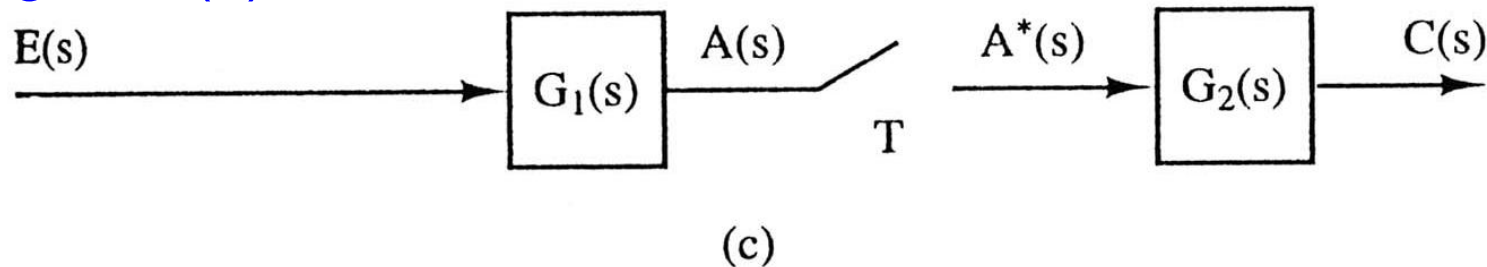
$$\overline{G_1 G_2}(z) = \mathcal{Z}[G_1(s)G_2(s)] \quad (4-18)$$

The bar above a product term indicates that the product must be performed in the s -domain before the z -transform is taken. In addition, note that

$$\overline{G_1 G_2}(z) \neq G_1(z)G_2(z) \quad (4-19)$$

that is, the z -transform of a product of functions is not equal to the product of the z -transforms of the functions.

Fig. 4.3 (c)



For the system of Figure 4-3c,

$$C(s) = G_2(s)A^*(s) = G_2(s)\overline{G_1 E^*}(s)$$

Thus

$$\underline{C(z) = G_2(z)\overline{G_1 E}(z)} \quad (4-20)$$

For this system a transfer function cannot be written; that is, we cannot factor $E(z)$ from $\overline{G_1 E}(z)$. $E(z)$ contains the values of $e(t)$ only at $t = kT$. But the signal $a(t)$ in Figure 4-3c is a function of all previous values of $e(t)$, not just the values at sampling instants. Since

$$A(s) = G_1(s)E(s)$$

then, from the convolution property of the Laplace transform,

$$a(t) = \int_0^t g_1(t - \tau)e(\tau) d\tau \quad (4-21)$$

Summary

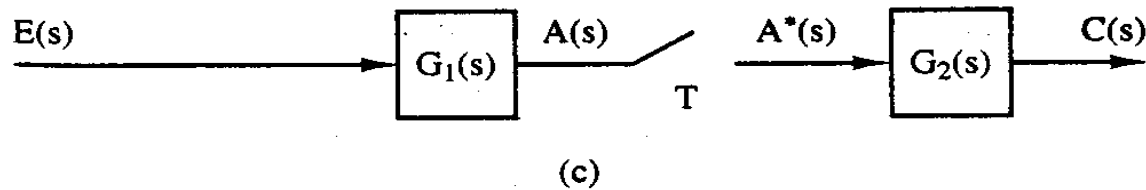
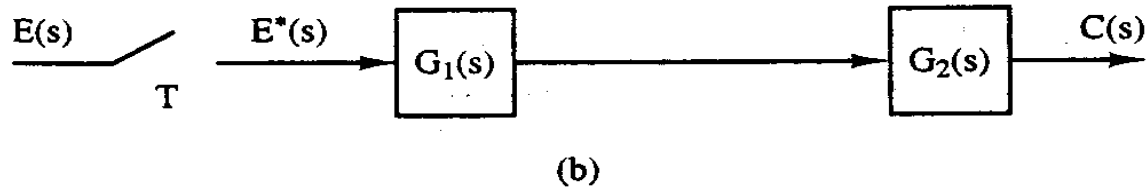
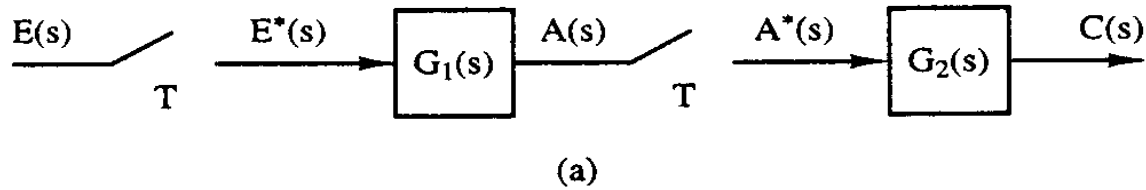


Figure 4-3 Open-loop sampled-data systems.

For 4.3(a) $C(z) = G_2(s)A^*(s)$ $A(z) = G_1(z)E(z)$ then $C(z) = G_1(z)G_2(z)E(z)$

For 4.3(b) $G_2(s)$ Would not contain a data-hold transfer function

$$C(z) = G_1(s)G_2(s)E^*(s) \quad \underline{C(z) = \overline{G_1}G_2(z)E(z)} \quad \overline{G_1}G_2(z) \neq G_1(z)G_2(z)$$

For 4.3(c) $C(s) = G_2(s)A^*(s) = G_2(s)\overline{G_1}E^*(s)$ $C(z) = G_2(z)\overline{G_1}E(z)$

4.4 Open-Loop Systems Containing Digital Filters

4.4 OPEN-LOOP SYSTEMS CONTAINING DIGITAL FILTERS

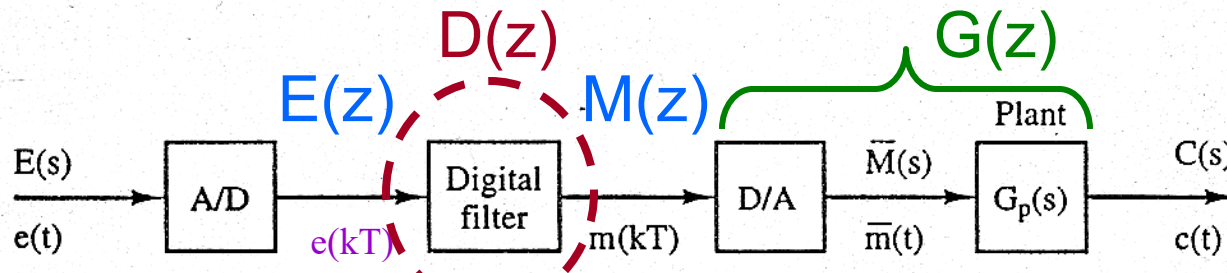


Figure 4-4 Open-loop system with a digital filter.

such that

$$M(z) = D(z)E(z) \quad (4-22)$$

or, through the substitution $z = e^{sT}$,

$$M^*(s) = D^*(s)E^*(s)$$

The D/A converter usually has an output data-hold register which gives the D/A the characteristics of a zero-order hold, and $\bar{m}(t)$ is of the form of $\bar{e}(t)$ in Figure 3-3. Thus the Laplace transform of the signal $\bar{m}(t)$ can be expressed as, from (3-2),

$$\bar{M}(s) = \frac{1 - e^{-Ts}}{s} M^*(s)$$

Hence

$$C(s) = G_p(s)\overline{M}(s) = G_p(s)\frac{1 - e^{-Ts}}{s}M^*(s)$$

Then, from (4-22),

$$C(s) = G_p(s)\frac{1 - e^{-Ts}}{s}D(z)|_{z=e^{Ts}}E^*(s) \quad (4-23)$$

and we see that the filter and associated A/D and D/A converters can be represented in block-diagram form as shown in Figure 4-5. Hence, from (4-23) or Figure 4-5,

$$C(z) = \mathcal{Z}\left[G_p(s)\frac{1 - e^{-Ts}}{s}\right]D(z)E(z) = G(z)D(z)E(z) \quad (4-24)$$

The digital computing device which implements the digital filter in Figure 4-4 actually processes the values of the input data samples $\{e(kT)\}$. However, our model for the digital filter processes a sequence of *impulse functions* of weight $\{e(kT)\}$. Hence the complete model must be used as depicted in Figure 4-5; that is, the combination of an ideal sampler, $D(z)$, and a zero-order hold does accurately model the combination of the A/D, digital filter, and D/A.

Summary

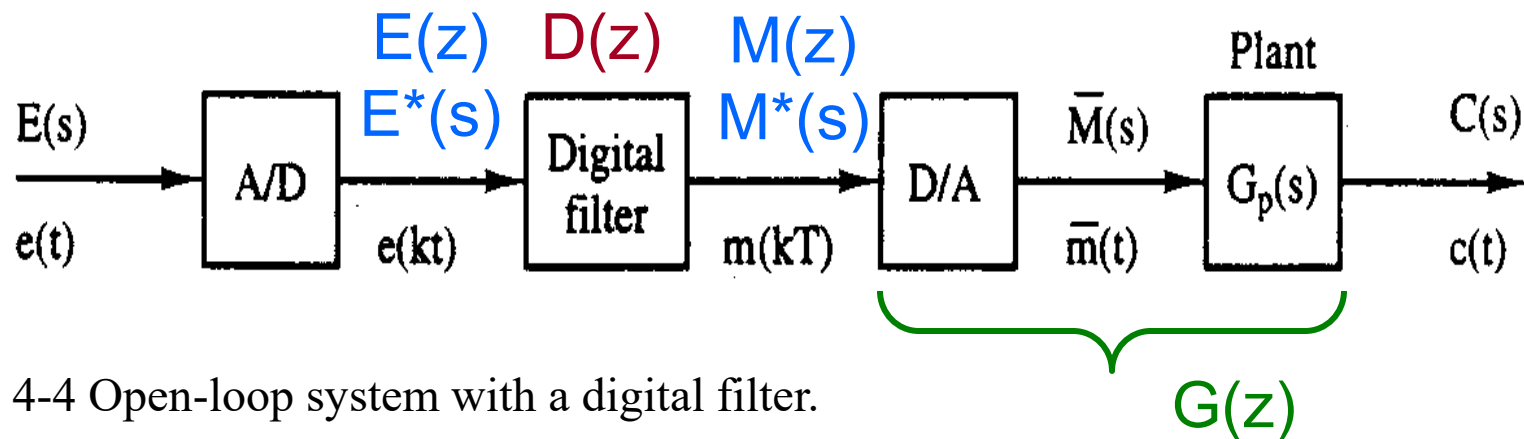


Figure 4-4 Open-loop system with a digital filter.

Such that
$$M(z) = D(z)E(z)$$

$$\bar{M}(s) = \frac{1 - \varepsilon^{-Ts}}{s} M^*(s)$$

then
$$C(s) = G_p(s) \frac{1 - \varepsilon^{-Ts}}{s} D(z) \Big|_{z=\varepsilon^{Ts}} E^*(s)$$

$$C(z) = z \left[G_p(s) \frac{1 - \varepsilon^{-Ts}}{s} \right] D(z) E(z) = G(z) D(z) E(z)$$

Example 4.4

Let us determine the step response of the system shown in Figure 4-5. Suppose that the filter is described by the difference equation

$$m(kT) = 2e(kT) - e[(k-1)T] \quad \text{Digital filter}$$

and thus

$$D(z) = \frac{M(z)}{E(z)} = 2 - z^{-1} = \frac{2z - 1}{z}$$

In addition, suppose that

$$G_p(s) = \frac{1}{s+1}$$

Then, as shown in Example 4.3,

$$\mathcal{Z}\left[\frac{1 - e^{-Ts}}{s(s+1)}\right] = \frac{1 - e^{-T}}{z - e^{-T}}$$

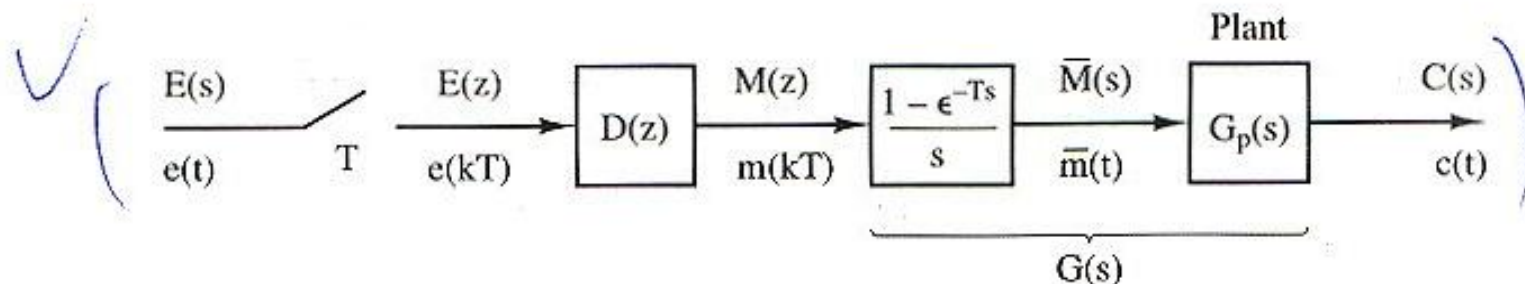


Figure 4-5 Model for the open-loop system.

Since $E(z) = z/(z - 1)$, from (4-24) we obtain

$$\begin{aligned} C(z) &= D(z)G(z)E(z) \\ &= \frac{2z - 1}{z} \left[\frac{1 - \epsilon^{-T}}{z - \epsilon^{-T}} \right] \frac{z}{z - 1} = \frac{(2z - 1)(1 - \epsilon^{-T})}{(z - 1)(z - \epsilon^{-T})} \end{aligned}$$

By partial fractions,

$$\frac{C(z)}{z} = \frac{(2z - 1)(1 - \epsilon^{-T})}{z(z - 1)(z - \epsilon^{-T})} = \frac{1 - \epsilon^T}{z} + \frac{1}{z - 1} + \frac{\epsilon^T - 2}{z - \epsilon^{-T}}$$

and

$$C(z) = (1 - \epsilon^T) + \frac{z}{z - 1} + \frac{(\epsilon^T - 2)z}{z - \epsilon^{-T}}$$

Since

$$\mathcal{Z}^{-1}[k_i z^{-i}] = \begin{cases} k_i, & n = i \\ 0, & n \neq i \end{cases}$$

the inverse z -transform of the first term in $C(z)$ has a nonzero value only for $n = 0$, and thus

Step resp. $\left\{ \begin{aligned} &\underline{c(nT) = 1 + (\epsilon^T - 2)\epsilon^{-nT}}, \quad n = 1, 2, 3, \dots \\ &c(0) = 1 - \epsilon^T + 1 + \epsilon^T - 2 = 0 \end{aligned} \right.$

Sol.之另一表示:

$$c(nT) = (1 - \epsilon^T)\delta(n) + 1 + (\epsilon^T - 2)\epsilon^{-nT}$$

Note that the value $c(0) = 0$ is obvious from inspection of $C(z)$, since the order of the numerator is less than the order of the denominator. The verification of this example is considered in Problem 4-11.

We can partially verify the response of the system of Example 4.4. Note that the final value of $c(nT)$ is unity. From the final value property of the z -transform, we verify this value:

$$\lim_{n \rightarrow \infty} c(nT) = \lim_{z \rightarrow 1} (z - 1)C(z) = \lim_{z \rightarrow 1} \frac{(2z - 1)(1 - e^{-T})}{z - e^{-T}} = 1$$

In addition, from (4-14), the dc gain of the system is given by

$$\text{dc gain} = D(z) \Big|_{z=1} G_p(s) \Big|_{s=0} = \frac{2z - 1}{z} \Big|_{z=1} \frac{1}{s + 1} \Big|_{s=0} = 1$$

Since in the steady state, the input is constant at a value of unity, the steady-state value of the output is

$$\lim_{n \rightarrow \infty} c(nT) = (\text{dc gain})(\text{constant input}) = (1)(1) = 1$$

which once again verifies the steady-state output.

As a final point in this example, note that the output of the digital filter is given by

$$M(z) = (2 - z^{-1})E(z) \Rightarrow m(kT) = 2u(kT) - u[(k - 1)T]$$

➡ Steady-state output of unity.

4.5 The Modified z-transform

Ideal time delays: $\Delta T, 0 < \Delta \leq 1$

Ordinary z-transform of the delayed time function is

$$Z[e(t - \Delta T)u(t - \Delta T)] = Z\left[E(s)\varepsilon^{-\Delta Ts}\right] = \sum_{n=1}^{\infty} e(nT - \Delta T)Z^{-n} \quad (4-25)$$

Delayed z-transform
(4-26)

$$E(z, \Delta) = z[e(t - \Delta T)u(t - \Delta T)] = z\left[E(s)\varepsilon^{-\Delta Ts}\right]$$

Example 4.5

Find $E(z, \Delta)$, if $\Delta = 0.4$, for $e(t) = \epsilon^{-at} u(t)$. From (4-25),

$$\begin{aligned} E(z, \Delta) &= \epsilon^{-0.6aT} z^{-1} + \epsilon^{-1.6aT} z^{-2} + \epsilon^{-2.6aT} z^{-3} + \dots \\ &= \epsilon^{-0.6aT} z^{-1} [1 + \epsilon^{-aT} z^{-1} + \epsilon^{-2aT} z^{-2} + \dots] \\ &= \frac{\epsilon^{-0.6aT} z^{-1}}{1 - \epsilon^{-aT} z^{-1}} = \frac{\epsilon^{-0.6aT}}{z - \epsilon^{-aT}} \end{aligned}$$

The sketches in Figure 4-6 show both $e(t)$ and $e(t - \Delta T)$.

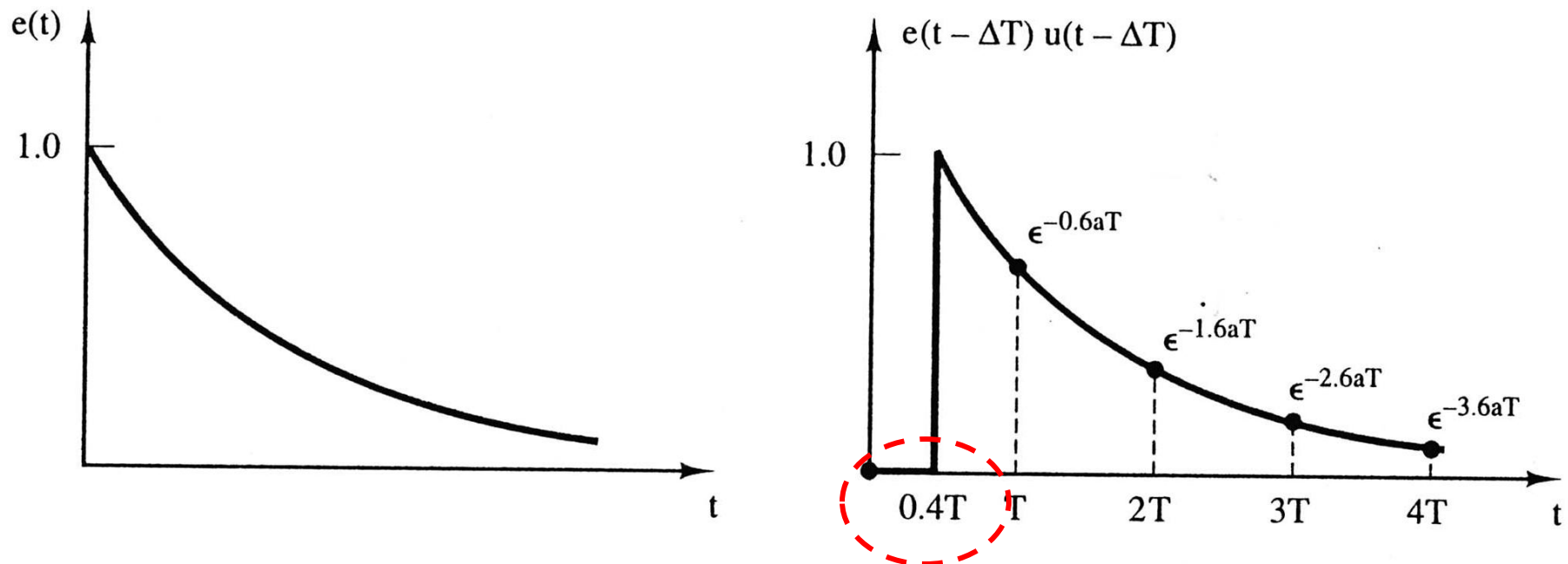


Figure 4-6 Example of the delayed z-transform.

Delayed z-transform
(4-26)

$$E(z, \Delta) = z \left[e(t - \Delta T) u(t - \Delta T) \right] = z \left[E(s) \varepsilon^{-\Delta T s} \right]$$

Modified z-transform
(4-27)

$$E(z, m) = E(z, \Delta) \Big|_{\underline{\Delta = 1 - m}} = z \left[E(s) \varepsilon^{-\Delta T s} \right] \Big|_{\Delta = 1 - m}$$

➡ Appendix VIII (the rightest column)

Two properties of the modified z-transform are

$$E(z, 1) = E(z, m) \Big|_{m=1} = E(z) - e(0) \quad \text{No delay} \quad (4-29, 30)$$

$$E(z, 0) = E(z, m) \Big|_{m=0} = z^{-1} E(z) \quad \text{A delay of one sample period, } T$$

Example 4.6

It is desired to find the modified z-transform of $e(t) = e^{-t}$. From (4-28),

$$\begin{aligned} E(z, m) &= e^{-mT} z^{-1} + e^{-(1+m)T} z^{-2} + e^{-(2+m)T} z^{-3} + \dots \\ &= e^{-mT} z^{-1} [1 + e^{-T} z^{-1} + e^{-2T} z^{-2} + \dots] \\ &= \frac{e^{-mT} z^{-1}}{1 - e^{-T} z^{-1}} = \frac{e^{-mT}}{z - e^{-T}} \end{aligned}$$

A sketch of the time functions is given in Figure 4-7.

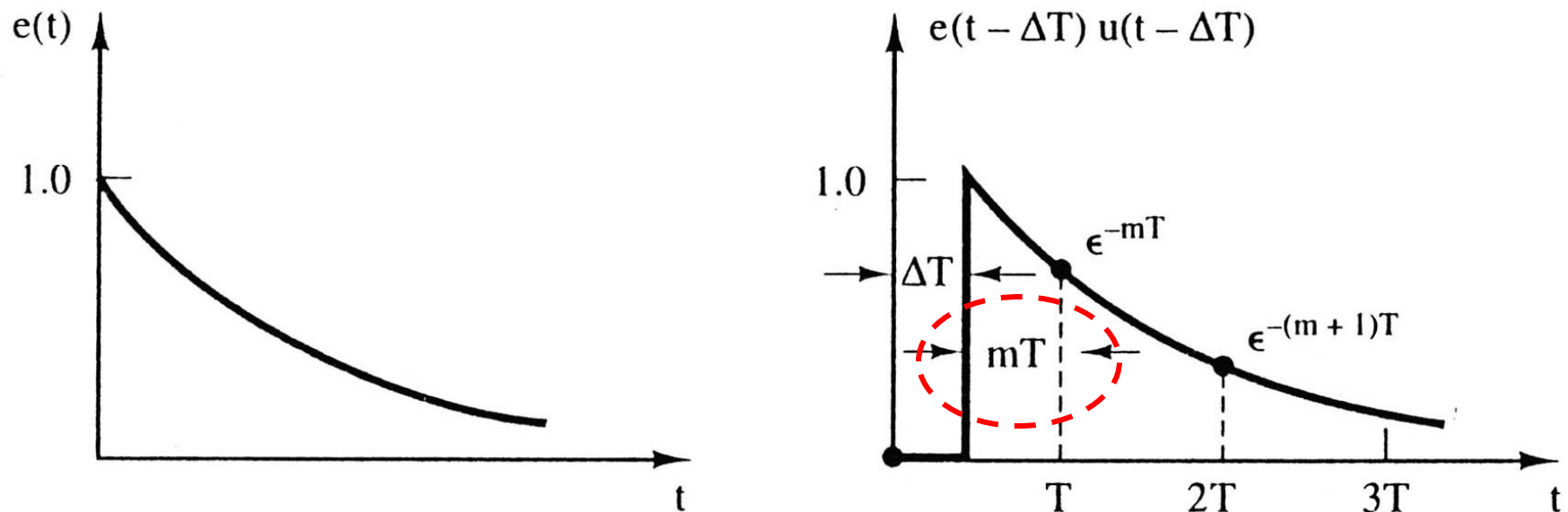


Figure 4-7 Example of the modified z-transform.

4.6 Systems with Time Delays

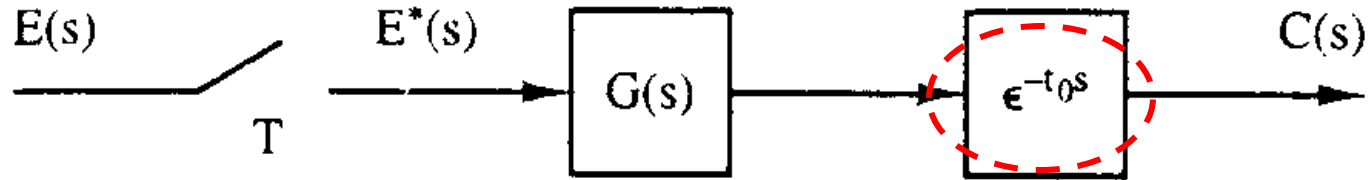


Figure 4-8 System with ideal time delay.

$$C(s) = G(s) \varepsilon^{-t_0 s} E^*(s) \quad \text{Ideal time delay of } t_0 \text{ sec.}$$

$$C(z) = z \left[G(s) \varepsilon^{-t_0 s} \right] E(z)$$

k is a positive integer

$$\text{Let } t_0 = \underline{kT} + \Delta T, \quad 0 < \Delta < 1$$

$$m = 1 - \Delta$$

$$C(z) = z^{-k} z \left[G(s) \varepsilon^{-\Delta Ts} \right] E(z) = \underline{z^{-k}} G(z, m) E(z)$$

Example 4.8

In Figure 4-8, let the input be a unit step, $t_0 = 0.4T$, and

$$G(s) = \frac{1 - e^{-Ts}}{s(s + 1)}$$

This system was considered in Example 4.3 with no delay. From (4-35) and the modified z-transform tables,

$$\begin{aligned} G(z, m) &= \mathcal{Z}_m \left[\frac{1 - e^{-Ts}}{s(s + 1)} \right] = (1 - z^{-1}) \mathcal{Z}_m \left[\frac{1}{s(s + 1)} \right] \\ &= \frac{z - 1}{z} \left[\frac{z(1 - e^{-mT}) + e^{-mT} - e^{-T}}{(z - 1)(z - e^{-T})} \right] \end{aligned}$$

Thus, since $mT = T - \Delta T = 0.6T$,

$$m = 1 - \Delta$$

$$G(z, m) = \frac{z - 1}{z} \left[\frac{z(1 - e^{-0.6T}) + e^{-0.6T} - e^{-T}}{(z - 1)(z - e^{-T})} \right]$$

Since $k = 0$ in (4-39), $C(z)$ is seen to be

$$C(z) = G(z, m) \frac{z}{z - 1} = \frac{z(1 - e^{-0.6T}) + e^{-0.6T} - e^{-T}}{(z - 1)(z - e^{-T})}$$

By the power-series method of Section 2.6,

$$C(z) = (1 - e^{-0.6T})z^{-1} + (1 - e^{-1.6T})z^{-2} + (1 - e^{-2.6T})z^{-3} + \dots$$

From Example 4.3, the response of this system with no delay is $c(nT) = 1 - e^{-nT}$. This response delayed by $0.4T$ is then

$$c(nT)|_{n \leftarrow (n - 0.4)} = 1 - e^{-(n - 0.4)T}, \quad n \geq 1 \quad (4-40)$$

which checks the results of this example.

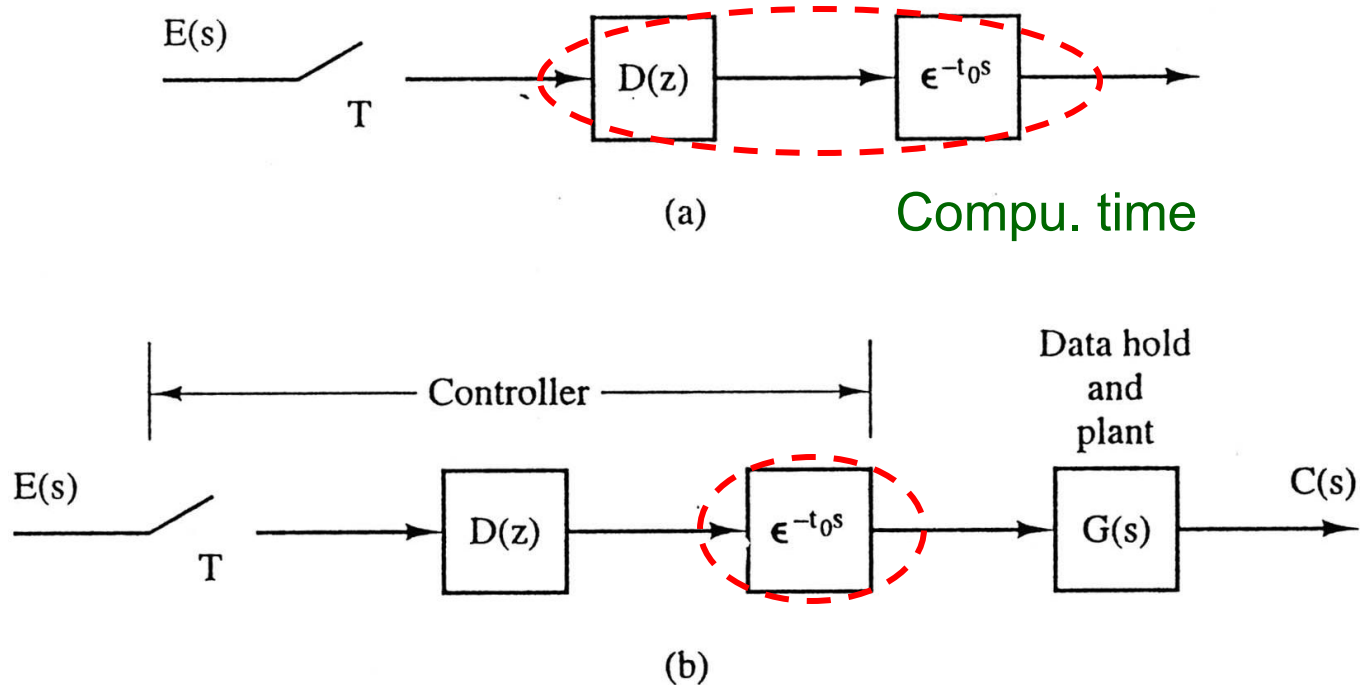


Figure 4-9 Digital controller with nonzero computation time.

Time Delay of Digital Controller

$$\begin{aligned}
 m(k) = & b_n e(k) + b_{n-1} e(k-1) + \cdots + b_0 e(k-n) \\
 & - a_{n-1} m(k-1) - \cdots - a_0 m(k-n)
 \end{aligned}
 \tag{4-41}$$

$$C(z) = z \left[G(s) \varepsilon^{-t_0 s} \right] D(z) E(z) \quad t_0 = \underline{kT} + \Delta T, \quad 0 < \Delta < 1$$

$$\Rightarrow \underline{C(z)} = \underline{z^{-k}} G(z, m) D(z) E(z) \tag{4-44}$$

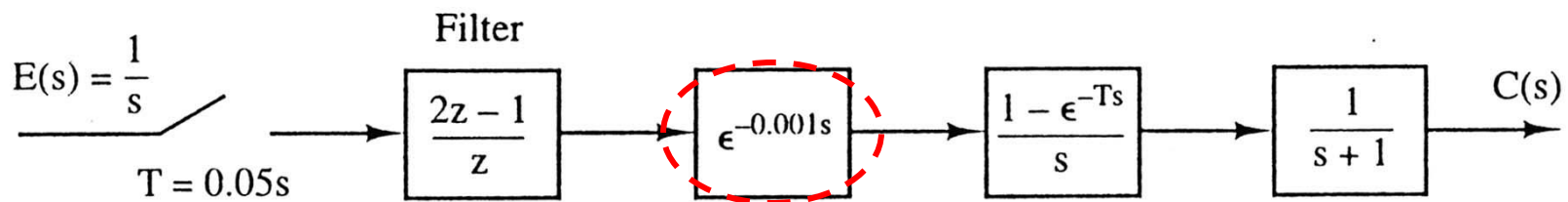


Figure 4-10 System for Example 4.9.

Example 4.9

Consider the system of Figure 4-10. This system is that of Example 4.4, with a computational delay added for the filter. The delay is 1 ms ($t_0 = 10^{-3}$ s) and $T = 0.05$ s. Thus, for this system,

$$D(z) = \frac{2z - 1}{z}$$

$$mT + \Delta T = T$$

or

$$mT = T - \Delta T = 0.05 - 0.001 = 0.049$$

$$m = 1 - \Delta$$

Then

$$G(z, m) = \mathcal{Z}_m \left[\frac{1 - e^{-Ts}}{s(s+1)} \right]_{mT=0.049} = \frac{z-1}{z} \mathcal{Z}_m \left[\frac{1}{s(s+1)} \right]_{mT=0.049}$$

From Example 4.8,

$$G(z, m) = \frac{z-1}{z} \left[\frac{z(1 - e^{-0.049}) + (e^{-0.049} - e^{-0.05})}{(z-1)(z - e^{-0.05})} \right]$$

Since the input is a unit step, then, from (4-44),

$$\begin{aligned} C(z) &= G(z, m)D(z)E(z) \\ &= \frac{z-1}{z} \left[\frac{z(1 - e^{-0.049}) + (e^{-0.049} - e^{-0.05})}{(z-1)(z - e^{-0.05})} \right] \frac{2z-1}{z} \left[\frac{z}{z-1} \right] \\ &= \frac{(2z-1)[z(1 - e^{-0.049}) + (e^{-0.049} - e^{-0.05})]}{z(z-1)(z - e^{-0.05})} \end{aligned}$$

4.7 Non-Synchronous Sampling

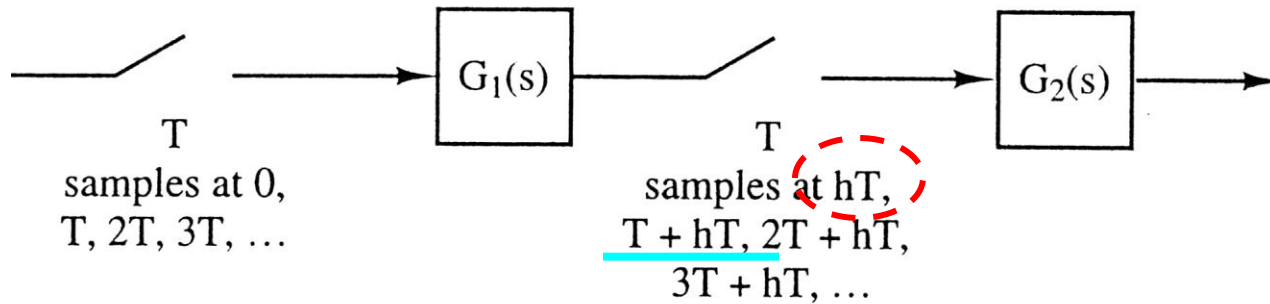


Figure 4-11 System with nonsynchronous sampling.

Both samplers operate at the same rate,
but are not synchronous.

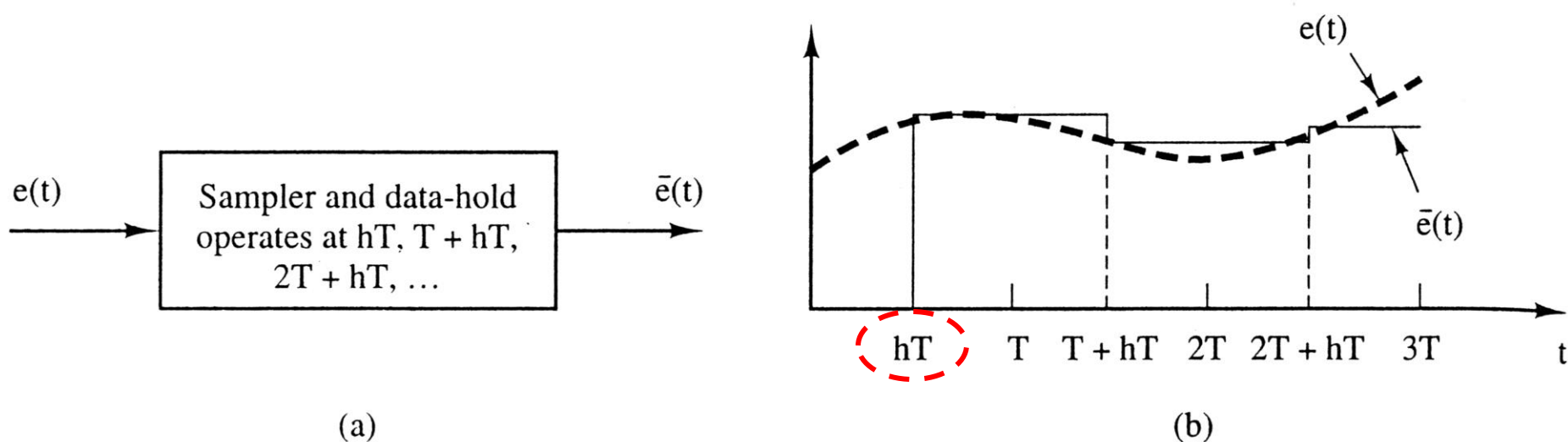


Figure 4-12 Illustration of nonsynchronous sampling.

$$\begin{aligned}
 \bar{e}(t) = & \underline{e(hT)}[u(t - hT) - u(t - T - hT)] + \underline{e(T + hT)}[u(t - T - hT) \\
 & - u(t - 2T - hT)] + e(2T + hT)[u(t - 2T - hT) \\
 & - u(t - 3T - hT)] + \dots
 \end{aligned}
 \tag{4-45}$$

Special case: $hT=0 \rightarrow$ Ch3 Eq. (3-1)

$$\bar{E}(s) = \frac{1 - \varepsilon^{-Ts}}{s} \varepsilon^{Ts} \varepsilon^{-hTs} E(z, m) \Big|_{m=h, z=\varepsilon^{Ts}} \quad (4-46)$$

$$E(z, m) = z \left[E(s) \varepsilon^{-\Delta Ts} \right]_{\Delta=1-m} \quad (4-47)$$

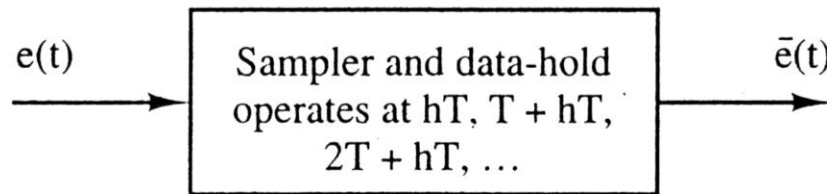
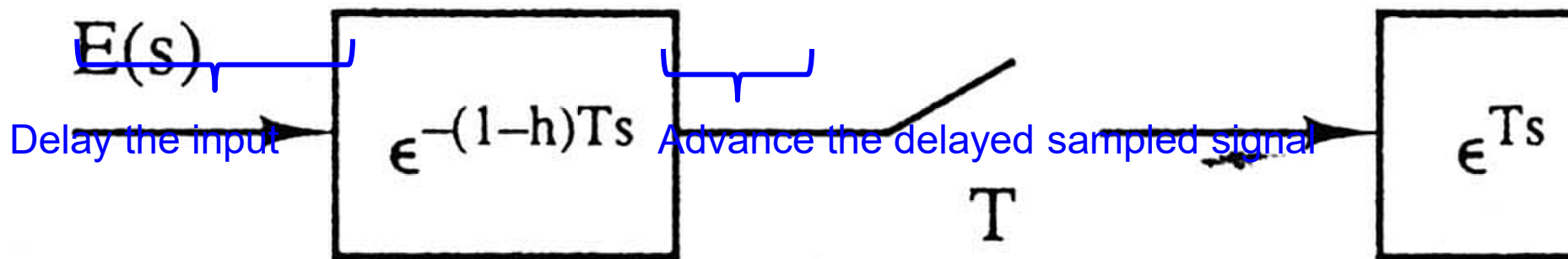


Fig. 4-12 (a))

$E(s)$

$\bar{E}(s)$



The total delay in the signal is zero.

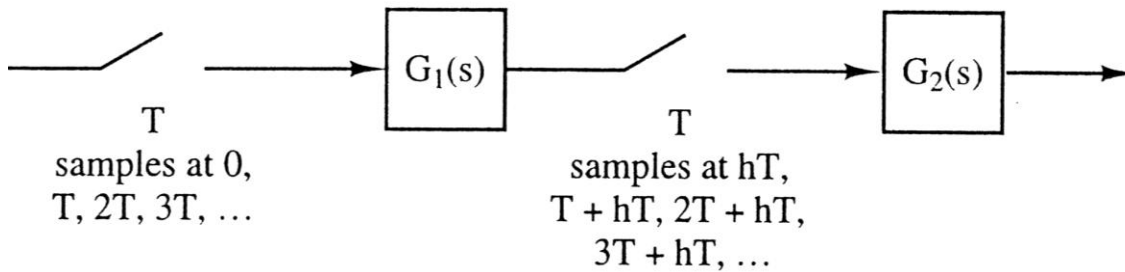


Figure 4-11 System with nonsynchronous sampling.

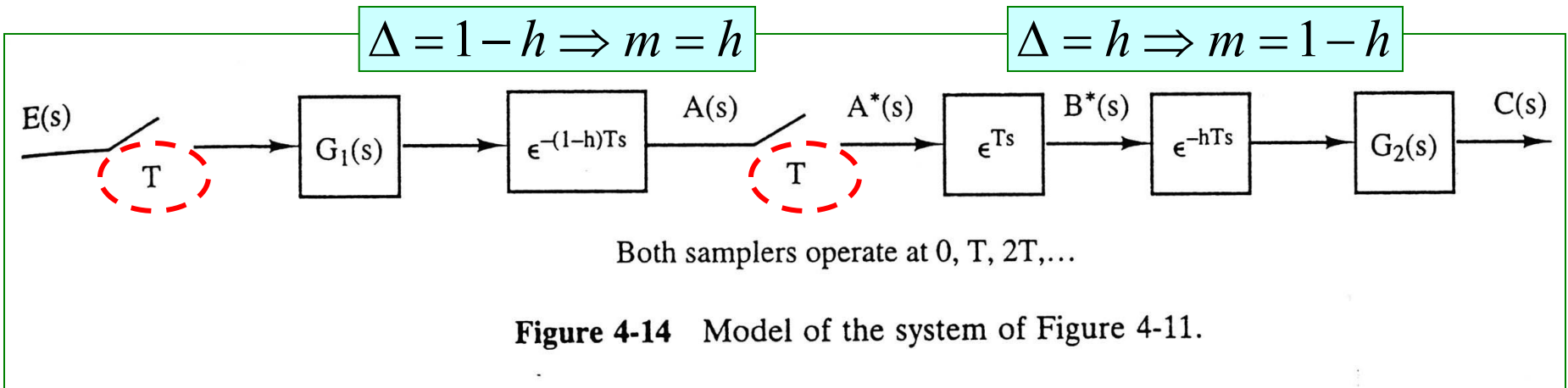


Figure 4-14 Model of the system of Figure 4-11.

$$\underline{A(z)} = G_1(z, m) \Big|_{m=h} E(z)$$

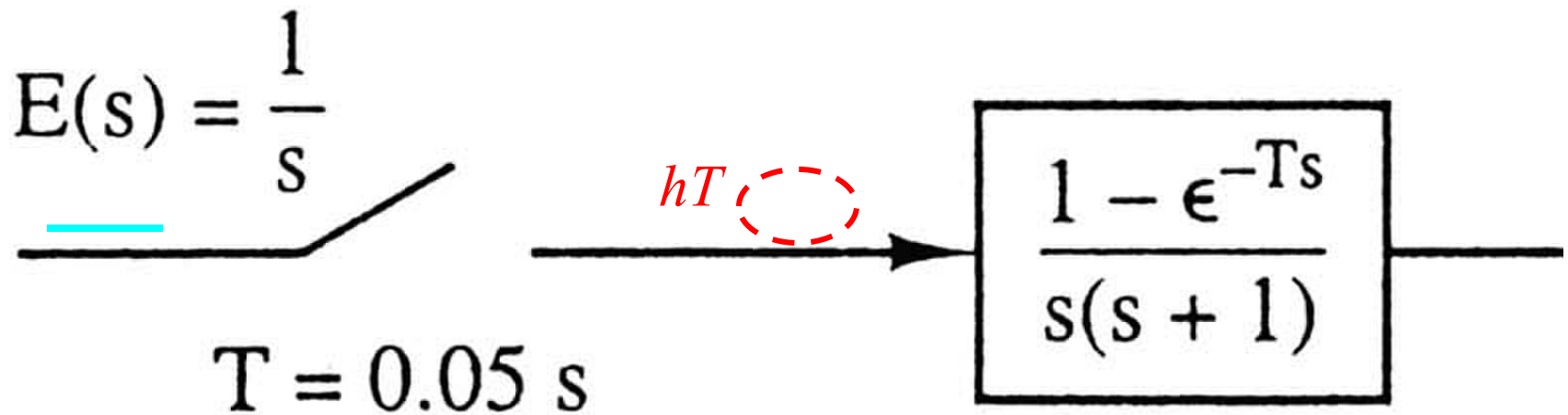
$$\underline{B(z)} = \underline{zA(z)} = zE(z)G_1(z, m) \Big|_{m=h}$$

$$C(z) = G_2(z, m) \Big|_{m=1-h} \underline{B(z)}$$

$$\Rightarrow \underline{C(z) = zE(z)G_1(z, m) \Big|_{m=h} G_2(z, m) \Big|_{m=1-h}}$$

Example 4.10

Non-Synchronous sampling



We wish to find $C(z)$ for the system of Figure 4-15, which contains nonsynchronous sampling. Now

$$\underline{E(z)} = \mathcal{Z}\left[\frac{1}{s}\right] = \frac{z}{z - 1}$$

For the system, $T = 0.05$ and $hT = 0.01$. From Example 4.8,

$$\underline{G_1(z, m)} = \mathcal{Z}_m\left[\frac{1 - e^{-Ts}}{s(s + 1)}\right] = \frac{z - 1}{z} \left[\frac{z(1 - e^{-mT}) + e^{-mT} - e^{-T}}{(z - 1)(z - e^{-T})} \right]$$

Then

$$G_1(z, m)|_{mT=0.01} = \frac{z-1}{z} \left[\frac{z(1 - \epsilon^{-0.01}) + \epsilon^{-0.01} - \epsilon^{-0.05}}{(z-1)(z - \epsilon^{-0.05})} \right]$$

Also, from the modified z -transform tables,

$$G_2(z, m) = \mathcal{Z}_m \left[\frac{1 - \epsilon^{-Ts}}{s^2} \right] = \frac{z-1}{z} \left[\frac{mTz - mT + T}{(z-1)^2} \right]$$

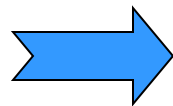
or

$$G_2(z, m)|_{m=1-h} = \frac{0.04z + 0.01}{z(z-1)}$$

Then, from the development above and (4-51),

$$\begin{aligned} C(z) &= z \left[\frac{z}{z-1} \right] \frac{z-1}{z} \left[\frac{z(1 - \epsilon^{-0.01}) + \epsilon^{-0.01} - \epsilon^{-0.05}}{(z-1)(z - \epsilon^{-0.05})} \right] \frac{0.04z + 0.01}{z(z-1)} \\ &= \frac{(0.04z + 0.01)[z(1 - \epsilon^{-0.01}) + \epsilon^{-0.01} - \epsilon^{-0.05}]}{(z-1)^2(z - \epsilon^{-0.05})} \end{aligned}$$

- See Summary (P161)
- Problems: 4-1, 4-2, 4-7, 4-10(a)
4-13, 4-19(a), 4-20(a)



Test Bank